Practical Curriculum Design: Security, Automation, & IoT Systems

Total Course Duration: 16 Weeks (80 Days)

Time Commitment: 5 Days/Week, 2 Hours/Day (Total: 160 Hours)

Delivery Model: Practical-Oriented (30% Concepts/Theory, 70% Hands-On Labs)

1. Overall Module Weightage & Time Allocation

To ensure students are industry-ready, the curriculum is divided into 4 Core Modules plus dedicated capstone Practical Labs. The time allocation is weighted based on the complexity of installation, programming requirements, and market demand.

Module	Duration	Total Hours	Weightage (%)	Primary Focus
Module 1: Electrical & Networking	3 Weeks	30 Hours	18.75%	Safe power wiring, network configuration, cabling
Module 2: Professional CCTV Systems	3.5 Weeks	35 Hours	21.875%	Camera selection, storage design, analytics, IP configuration
Module 3: Security & Building Automation	3.5 Weeks	35 Hours	21.875%	Commercial security systems, fire alarms, access control
Module 4: Smart Home, IoT & Business	4 Weeks	40 Hours	25.00%	Home protocols, system integration, entrepreneurship, sales
Professional Practical Labs (Capstone)	2 Weeks	20 Hours	12.50%	Integrated system deployment, splicing, troubleshooting
Total	**16 Weeks**	**160 Hours**	**100%**	**Comprehensive practical competency**

2. Detailed Topic Weightage & Hours Breakdown

Below is the recommended weightage for each topic within the modules, structured as compact 7-column tables indicating weeks (Wk), hours (Hrs), sessions (Sess), focus, materials required, and practical session works.

Module 1: Electrical & Networking (30 Hours)

This module establishes the foundational hardware and network infrastructure.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W1	**Electrical Fundamentals (AC/DC)**	2 Hrs	1.0	- Concept of voltage and current - Resistance basics - Series/parallel loops & two-way switch logic	- Modular switch & 3-pin socket - Two-way SPDT switches & bulb - 1.5 sq mm copper wire - Digital DMM	- Mount switch/socket on board - Wire live/neutral loop connections - Wire two-way switch staircase loop

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W1	**Earthing & Lightning Protection**	2 Hrs	1.0	- Earth pit low resistance path - Chemical vs salt-charcoal pits - Surge Protection Devices (SPDs)	- GI earth pipe & grounding wire - Brass connection bolts - Salt, charcoal, sand backfill - Series test lamp	- Connect ground wire clamp to pipe - Layout backfill layers in test pit - Perform Phase-to-Earth bulb glow test
W1	**Multimeter Practical**	3 Hrs	1.5	- AC/DC voltage dial settings - Continuity test limits - Insulation resistance checks	- Multimeter & probes - Test lamp - Junction box with simulated faults	- Measure Phase-Neutral/Phase-Earth/Neutral-Earth - Use continuity mode to find broken cores - Hunt for short-circuit faults inside box
W1	**Electrical Safety & PPE**	1 Hr	0.5	- Shock current threshold levels - LOTO isolation guidelines - Safety breaker topologies	- Single-phase DB board - double-pole MCB breaker - 30mA RCCB leakage safety - 60W test lamp & holder	- Install MCB & RCCB inside DB - Wire switchboard through breakers - Connect test lamp Phase-to-Earth to trip RCCB
W1/2	**Power Supply & SMPS**	2 Hrs	1.0	- Linear vs SMPS conversion - Overload capacity safety margins - Battery backup sizing calculations	- 12V 10A CCTV SMPS unit - Multi-output block - 12V 7Ah SLA battery - Digital multimeter	- Mount and wire SMPS input terminals - Adjust output pot to 12.2V DC - Connect backup battery and test load runtime
W1/2	**Reading Electrical Drawings**	2 Hrs	1.0	- Standard symbols & grid coordinates - Relays (SPDT/DPDT) & 0/1 signal logic - Switching triggers (Dry vs Wet contacts)	- Site wiring schematics & relay board - 12V EM lock & exit push button - Wire ferrule tags & crimper	- Trace active paths on 3-page schematic - Wire relay door lock loops matching prints - Test fire panels emergency cut-off trip
W1/2	**Cable Selection & Management**	2 Hrs	1.0	- AWG vs metric sq mm sizing - Line voltage drop calculations - Conduit 40% fill rule	- PVC conduits, spacer clips, plugs - Manual bending spring - Binding wire & junction boxes - PVC insulation tape	- Mount PVC conduits onto board - Bend 90° curves using spring - Pull wires using binding wire - Make pigtail joints inside box & tape-wrap
W2	**IP Addressing & Subnetting**	4 Hrs	2.0	- IPv4 subnet CIDR masks (/24, /28) - Default gateways & static IPs - Private RFC 1918 ranges	- Laptops (Windows) - Cat6 network patch cords - 8-port network switch - Marker board	- Set up static IPs on two laptops - Verify end-to-end ping connection - Diagnose and resolve wrong gateway settings
W2/3	**Router & Switch Configuration**	4 Hrs	2.0	- WAN vs LAN interface setup - NAT & Port forwarding rules - Wi-Fi SSID and WPA2 setup	- Dual-band Wi-Fi router - 8-port network switch - Ethernet cables & laptops	- Access router administrator panel - Configure IP pool and reservation - Set up port forwarding for external cameras - Calibrate Wi-Fi channel spacing
W3	**VLAN Basics**	2 Hrs	1.0	- VLAN broadcast segment separation - IEEE 802.1Q tagging standards - Access vs Trunk port routing	- Layer 2 Managed switch - Internet router - Laptops & patch cords	- Access managed switch GUI console - Create VLAN 10 (Data) & VLAN 20 (CCTV) - Assign specific access ports & configure trunk link

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W3	**PoE & PoE Switches**	2 Hrs	1.0	- 802.3af/at/bt wattage levels - Active vs Passive PoE - Switch power budgets	- PoE Switch - PoE IP camera & non-PoE camera - PoE splitter - Multimeter	- Measure active PoE pin voltages - Calculate total camera load budget - Wire non-PoE camera using splitter
W3	**Fiber Optic Basics**	2 Hrs	1.0	- Single-mode vs Multi-mode fiber - SFP transceiver standard speeds - Optical media converter setup	- SC-LC fiber patch cords - Fiber media converters - Gigabit switches with SFP ports - SFP modules	- Install SFPs in switch ports - Link switches using fiber patch cable - Verify fiber link status on media converters
W3	**Network Troubleshooting**	2 Hrs	1.0	- RJ45 (T568A/B) & RJ11 standards - Diagnostic ping/tracert tools - STP loop & IP clash resolution	- RJ45/RJ11 plugs & Cat6/Phone cables - Crimper tools & LAN wiremap tester - Workstations & managed switch	- Crimp Cat6 (RJ45) & phone (RJ11) plugs - Run CMD pings/tracert link audits - Troubleshoot and resolve switch loops

Module 2: Professional CCTV Systems (35 Hours)

CCTV is one of the highest-demand skills. This module balances video analytics design with hardware setup.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W4	**Analog & IP CCTV**	2 Hrs	1.0	- Coaxial vs Cat6 layouts - HD-TVI analog protocols - Video balun impedance matching	- HD-TVI Analog camera & IP Dome - Coaxial cable & Cat6 spool - BNC connectors & video baluns	- Crimp BNC connector on coaxial cable - Wire video baluns over Cat6 runs - Connect feeds to DVR/NVR to compare latency
W4	**Camera Selection & Design**	3 Hrs	1.5	- Camera form factors (Dome, Bullet) - Low-light lux ratings - DORI standard guidelines	- Dome, Bullet, and Turret cameras - Wall bracket mounts - Lux level datasheets	- Mount Dome and Bullet cameras on brackets - Verify WDR backlight compensation settings - Test camera output under dark/low lux settings
W4	**Lens & Field of View**	2 Hrs	1.0	- Focal length vs angle of view - Fixed vs varifocal lenses - Lens calculation software	- Fixed cameras (2.8mm & 4mm) - Motorized varifocal camera - Lens calculator mobile application	- Compare FOV angles of 2.8mm and 4mm lenses - Calibrate varifocal zoom/focus remotely - Run lens calculations for a mock warehouse
W4	**DVR, NVR & Hybrid Systems**	3 Hrs	1.5	- NVR vs DVR architecture - SATA storage interfaces - PoE ports integration	- 8-ch NVR & 8-ch DVR - 1TB Surveillance HDD - SATA cables & screwdriver	- Install hard drive inside NVR casing - Connect SATA power and data cables - Initialize HDD and configure recording profiles
W5	**RAID & Storage Design**	2 Hrs	1.0	- RAID 0, 1, 5, 10 redundancy - Storage retention calculators - surveillance vs desktop HDDs	- Laptops - Storage calculator tool - Multi-bay test NVR	- Set up a RAID 1 backup array inside NVR - Calculate storage storage for 10 IP cameras - Test hot-swapping a HDD in RAID array
W5	**ONVIF Standards**	1 Hr	0.5	- Profile S, G, and T differences - Cross-brand device compatibility - ONVIF discovery protocols	- Hikvision NVR & Dahua IP camera - POE switch & patch cords	- Link Dahua IP camera to Hikvision NVR - Configure connection via ONVIF Profile S - Test video feed and PTZ command link

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W5	**Video Compression (H.264/H.265/H.265+)**	2 Hrs	1.0	- H.264 vs H.265 vs H.265+ codecs - CBR vs VBR bandwidth control - FPS impact on storage	- IP camera & NVR - Network bandwidth monitoring app	- Change camera compression profile to H.265+ - Measure live bandwidth drop compared to H.264 - Configure variable bitrate threshold settings
W5	**AI Analytics**	2 Hrs	1.0	- Line crossing rules setup - Intrusion detection parameters - human vs vehicle classification	- Smart IP camera & NVR - Laptop client	- Define line-crossing parameters on NVR - Configure intrusion detection alert zone - Test and record human/vehicle alarm filter
W5	**ANPR (Automatic Number Plate Recognition)**	2 Hrs	1.0	- ANPR camera mounting angles - Shutter speed configuration - Whitelist database triggers	- ANPR camera & NVR - Test vehicle license plates - Laptop client	- Adjust camera shutter speed to 1/1000s - Align camera angle within 30° limit - Upload whitelist database to trigger relay lock
W5/6	**Facial Recognition**	2 Hrs	1.0	- Face pixels enrollment criteria - Matching similarity thresholds - Database blacklisting rules	- Face recognition camera & NVR - Face database photos - Laptop client	- Enroll student faces into NVR database - Configure facial similarity limit to 85% - Test blacklist matching alert sound trigger
W6	**ColorVu, AcuSense & Thermal Cameras**	3 Hrs	1.5	- ColorVu supplemental lighting - AcuSense classification logic - Thermal fire point detection	- ColorVu camera & AcuSense camera - Handheld heat source (solder iron)	- Configure ColorVu night warm light settings - Verify AcuSense false-alarm tree-branch filter - Monitor heat spot alert on thermal camera feed
W6	**PTZ Cameras**	2 Hrs	1.0	- PTZ pan/tilt mechanics - Preset coordinates setup - Auto-tracking logic	- PTZ Dome camera - Controller joystick - NVR	- Connect PTZ control via RS485/IP - Program 5 preset camera locations - Configure a continuous scanning patrol tour
W6	**Remote Viewing & Cloud**	3 Hrs	1.5	- Cloud P2P server connection - Dynamic DNS (DDNS) setup - Mobile client apps configurations	â€¢ NVR & internet router â€¢ Phone with Hik-Connect/DMSS app â€¢ Laptop client	- Link NVR to Cloud P2P portal - Scan QR code to add device on mobile app - Set up port-forwarding for DDNS connection
W6/7	**Cybersecurity**	2 Hrs	1.0	- Disabling default ports (80, 554) - Password complexity criteria - HTTPS certificate creation	- IP camera, NVR, laptop	- Modify default HTTP/RTSP ports on camera - Disable UPnP network settings on NVR - Install HTTPS security certificate on camera
W7	**Troubleshooting & Maintenance**	4 Hrs	2.0	- No-video loop isolation - Resolving IP address clashes - IR halo ring cleaning	- Multimeter, lens cleaning fluid - Faulty cameras - IP scanner software	- Troubleshoot camera offline fault on bench - Resolve duplicate IP address clashes - Perform lens cleaning to remove IR light halo

Module 3: Security & Building Automation (35 Hours)

This module covers commercial low-voltage (ELV) systems.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W7	**Video Door Phone**	2 Hrs	1.0	- 2-wire vs IP VDP layouts - Lock relay integrations - SIP apartment routing	- VDP outdoor call station & monitor - 12V solenoid lock - 4-core signal cable	- Wire outdoor call station to indoor monitor - Connect lock release relay to solenoid lock - Test voice/video communication & lock release
W7	**Access Control**	3 Hrs	1.5	- Wiegand controller connections - Card reader configurations - Exit push-button loops	- Access control panel - RFID card reader & exit button - 12V electromagnetic lock - 12V 5A power supply	- Wire RFID reader and exit button to controller - Terminate EM lock power supply lines to panel - Register RFID cards & test access control loop
W8	**Biometric Systems**	2 Hrs	1.0	- Fingerprint/Facial scanner metrics - Standalone vs Network panels - Access software log retrieval	- Standalone biometric reader - RFID cards & network switch - Laptop with management software	- Enroll fingerprints and faces on reader panel - Configure network IP settings on reader - Retrieve access database log onto laptop
W8	**Smart Locks**	2 Hrs	1.0	- EM locks fail-safe vs fail-secure - Lock mounting bracket sizing - Flyback protection diode loop	- EM lock & Drop-bolt lock - L/Z mounting brackets - 1N4007 flyback diode - Screwdriver set	- Mount L/Z bracket to mock door frame - Install EM lock on bracket plates - Terminate flyback diode across lock terminals
W8	**Boom Barrier**	2 Hrs	1.0	- Barrier motor balance springs - Loop detector vehicle loops - Photo-beam safety sensors	- Boom barrier gate unit - Loop detector card - Loop coil wire spool - Safety photo-beams	- Wire loop detector card to barrier board - Lay loop wire coil in test pattern on floor - Test barrier safety auto-reverse using loop
W8	**Gate Automation**	2 Hrs	1.0	- Sliding vs swing motors - Gear rack alignment rules - Remote control pairing	- Sliding gate motor & gear rack - Limit switch contacts - Remote transmitter keyfobs	- Align gear rack to sliding motor pinion - Wire magnetic limit switches to controller - Pair RF remote keyfobs to receiver board
W8/9	**Intruder Alarm Systems**	3 Hrs	1.5	- Hardwired EOL resistor loops - PIR motion detector zones - Arming profiles setup	- Alarm panel unit - PIR motion detector & door contacts - EOL resistors & warning siren	- Wire PIR sensor to alarm panel zone terminals - Connect EOL resistor to zone loop contact - Configure panel arm profiles & trigger alarm
W9	**Fire Alarm Systems**	4 Hrs	2.0	- Conventional vs addressable panels - Smoke & heat detector layouts - Trip integration relays	- Conventional fire alarm panel - Smoke detector & heat detector - Fire hooter & EOL capacitor - Test smoke aerosol spray	- Wire smoke detector in zone terminal loop - Install EOL resistor on final zone terminal - Test trigger using smoke aerosol spray
W9	**PA/BGM Systems**	2 Hrs	1.0	- 100V line power step-up/down - Speaker impedance matching - Microphone override priority	- 100V PA amplifier - Ceiling speaker with LMT - Paging microphone	- Connect ceiling speaker on 100V line tap - Calculate speaker wattage match to amp load - Test microphone voice override trigger
W9	**Intercom Systems**	2 Hrs	1.0	- PABX trunk/extension ports - IP PBX SIP routing loops - Extension programming codes	- Analog PABX unit (1x4) - Analog phone instruments - RJ11 cables & punch-down tool	- Terminate extension lines onto PABX board - Configure extension call group settings - Run extension call forwarding test drills

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W9/10	**Elevator Access Control**	2 Hrs	1.0	- Floor relay board connections - Restricted access mapping - Emergency override relays	- Access controller - 8-ch relay output board - LED indicators & laptop	- Wire relay outputs to simulate elevator floor buttons - Program access level restrictions in software - Trigger fire input loop to release all relays
W10	**Visitor Management**	2 Hrs	1.0	- Badging card printing setup - Barcode scanning checks - Database logging parameters	- PC with visitor software - Webcam & USB barcode scanner - Thermal badge printer	- Install visitor management desktop client - Configure webcam visitor photo capture - Generate and print barcode visitor pass
W10	**Hotel Lock Systems**	2 Hrs	1.0	- Mifare card encryption logic - Energy saver room switches - PMS software connectivity	- Hotel lock unit & card encoder - Mifare cards - Energy saver room switch	- Encode check-in card using hotel software - Verify card opens offline hotel door lock - Test energy saver switch room power activation
W10	**Parking Management**	2 Hrs	1.0	- Ticket dispenser systems - Loop card collector wiring - Workstation parking logic	- Ticket dispenser unit - Loop coil wire spool - Barrier gate cabinet - Printer	- Wire ticket dispenser card to barrier board - Simulate loop input to print parking ticket - Test ticket barcode verification loop
W10	**Basic Building Management Systems (BMS)**	3 Hrs	1.5	- DDC input/output configurations - Modbus/BACnet standard setups - Sensor signal calibrations	- DDC controller unit - Temperature sensor probe - Laptop with BMS software	- Terminate temp sensor to DDC analog input - Calibrate sensor voltage scaling parameters - Verify Modbus register map readouts on PC

Module 4: Smart Home, IoT & Business (40 Hours)

Smart Home integration requires high logical skill (programming systems), and establishing the business requires commercial execution.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W11	**KNX Fundamentals**	4 Hrs	2.0	- KNX bus TP1 green wiring - ETS system configurations - Group addressing principles	- KNX power supply & choke - KNX switch actuator module - KNX wall switch panel - Green bus wire & ETS laptop	- Wire KNX devices in standard bus loop - Configure group addresses inside ETS software - Program KNX switch to activate actuator light
W11	**Wi-Fi Smart Home**	2 Hrs	1.0	- Tuya gateway configurations - Neutral vs capacitor wiring - Router client limit troubleshooting	- Wi-Fi smart switch modules - Tuya Zigbee hub gateway - Smart plug socket & router	- Wire Tuya switch on test board setup - Link smart switch to Tuya mobile application - Configure voice linking to Google Assistant
W11	**Zigbee & Z-Wave**	3 Hrs	1.5	- Mesh network architectures - Coordinator vs router roles - Low power sensor sleep states	- Zigbee USB coordinator hub - Zigbee motion sensor & bulbs - Laptops	- Form a Zigbee mesh network via coordinator - Add Zigbee motion sensor and smart bulb - Test mesh self-healing by powering off router
W11/12	**Matter & Thread**	2 Hrs	1.0	- Matter unified standard layers - Thread border router mesh - Multi-admin pairing setups	- Apple HomePod Mini (Thread) - Matter smart plug module - Phone	- Connect Thread plug to HomePod border router - Pair device to Google Home using Matter code - Test control from iOS and Android devices

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W12	**Alexa & Google Home Integration**	2 Hrs	1.0	- Cloud-to-Cloud OAuth linking - Custom routine scripting - Voice profile calibrations	- Amazon Echo Dot speaker - Smart switches & phone	- Link smart home app skills to Amazon Alexa - Write a "Good Night" routine for switches - Calibrate voice commands trigger triggers
W12	**Home Assistant**	3 Hrs	1.5	- Local HAOS hub installations - HACS custom integrations - YAML script automation rules	- Raspberry Pi 4 (HAOS ready) - Zigbee sensors & router	- Set up Home Assistant configuration panel - Integrate Zigbee coordinator dongle - Write automation (turn on light when motion)
W12	**Smart Lighting & Curtain**	2 Hrs	1.0	- Phase dimming (leading/trailing) - Curtain motor track calibrations - Switch relay wiring	- Smart dimmer module & lamp - Curtain motor with track spool - Control panel switch	- Wire smart dimmer module to lamp load - Set curtain motor physical open/close limits - Program manual switch override controls
W12/13	**Smart Irrigation & Energy Monitoring**	2 Hrs	1.0	- Solenoid valve wiring loops - Non-invasive CT clamp setup - Energy dashboard settings	- 24VAC water solenoid valve - Smart current transformer clamp - Main DB panel board	- Wire water valve to smart switch relay board - Install current clamp on mains power line - Verify load metrics display on HA dashboard
W13	**Solar CCTV & IoT Fundamentals**	2 Hrs	1.0	- MPPT charge controller sizing - Battery depth of discharge calculations - 4G industrial router configurations	- Solar panel & MPPT controller - 12V LiFePO4 battery pack - 4G LTE SIM router	- Wire solar panel to MPPT charger terminals - Terminate charger line to LiFePO4 battery - Configure 4G SIM APN and test failover link
W13	**Site Survey & BOQ/Quotation**	3 Hrs	1.5	- Site surveys checklists design - BOQ template structures - Pricing markup margins	- Site survey templates sheets - Laser distance meter tape - Laptops with Excel	- Perform physical room site survey measurements - Map camera node locations on layout sheet - Generate formal BOQ with hardware margins
W13	**Project Planning & Installation Standards**	3 Hrs	1.5	- Conduit path planning - Height mounting standards - Cable labeling schemas	- Wire label printer & markers - PVC conduit conduits - Cable binders	- Draw conduit pathways layouts for villa plan - Print and apply wire tags matching drawings - Secure cables with velcro straps inside rack
W13/14	**Testing, Commissioning & AMC**	2 Hrs	1.0	- Loop testing logs design - Handover checklist templates - AMC contract structures	- Handover checklists forms - AMC contract templates	- Complete mock system handover checklist - Draft comprehensive system AMC proposal - Conduct client training walkthrough script
W14	**Customer Support & Sales/Negotiation**	2 Hrs	1.0	- SLA support ticketing guidelines - Handling pricing objections - Value-based sales pitches	- SLA ticket templates - Sales script worksheets	- Roleplay client sales negotiations on site - Design support SLA ticket priority grid - Practice pitching custom design advantages
W14	**Digital Marketing & Business Automation**	2 Hrs	1.0	- GBP local SEO optimization - WhatsApp catalog setups - Automated quick replies	- PC client & smartphone - WhatsApp Business profile - Google Business Profile	- Setup and optimize Google Business Profile - Create WhatsApp Business product catalogs - Write automated greeting/away replies templates
W14	**Business Setup & Entrepreneurship**	1 Hr	0.5	- Company registration laws - Distributor registrations - Markup margins calculations	- Dealer registration forms - Business plan template	- Complete dealer registration forms - Calculate project startup costs and cash flow - Outline tax licensing guidelines (GST/VAT)

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Materials Required	Practical Session Works
W14	**Placement Training & Career Prep**	4 Hrs	2.0	- Resume designs layouts - LinkedIn profile optimizations - Mock technical interviews	- Resume templates sheets - Job search check list	- Draft professional resume with ELV portfolio - Setup and optimize LinkedIn career profile - Conduct mock technical interview sessions

Professional Practical Labs (20 Hours)

Week	Lab Name	Hours	Sessions	Target Lab Activities	Materials Required	Practical Session Works
W15	**Network Rack Setup & Fiber Splicing**	2 Hrs	1.0	- Rack unit layout arrangements - Fiber core stripping & cleaving - Fusion splicing operations	- 9U wall-mount network rack - patch panels & Cat6 cable spool - Fiber splicer, cleaver, pigtails	- Mount patch panels inside 9U rack cabinet - Strip and cleave fiber core clean using tools - Perform fusion splice and check core DB loss
W15	**Complete Office CCTV Installation**	4 Hrs	2.0	- Coaxial/Cat6 routing runs - Camera hardware mountings - NVR network config	- 4 IP Dome & 4 analog cameras - NVR, DVR, Cat6 patch panel - BNC/RJ45 jacks, display screen	- Run cables from cameras to central DB rack - Mount cameras, terminate BNC/RJ45 ends - Power up NVR, initialize HDD, configure feeds
W15	**Villa Smart Home Installation**	4 Hrs	2.0	- Switches loops wiring - HAOS hub installations - Smart device integrations	- Raspberry Pi, switch modules - Dimmer modules, curtain track - Amazon Echo Dot speaker	- Wire switch board modules on wall board - Install HAOS on Pi, integrate switch relay - Configure Alexa voice routine triggers
W16	**Access Control & Video Door Phone Setup**	3 Hrs	1.5	- VDP wiring integrations - Access panel loop wirings - EM locks bracket setups	- Access board, RFID reader panel - EM lock, VDP monitor station - 1.5mm wire coils, screwdriver	- Connect outdoor call station to VDP monitor - Wire access panel release relay to EM lock - Register RFID cards & test access lock loop
W16	**Gate Automation & Boom Barrier Setup**	3 Hrs	1.5	- Gear rack pinion alignments - Loop coil floor layouts - RF remote key pairings	- Gate motor, gears, barrier cabinet - Loop detector card, loop wire - RF keyfobs remotes	- Align gear rack to sliding gate motor gears - Wire loop detector card to barrier board - Pair RF remote keyfob to receiver card
W16	**Fire & Intruder Alarm Installation**	2 Hrs	1.0	- Conventional fire loop wirings - Wireless sensor integrations - Siren alarms trigger check	- Fire panel, smoke detector - Alarm panel, PIR, door contacts - EOL resistors, hooter sounder	- Wire smoke detector in conventional loop - Integrate wireless PIR to alarm panel zones - Configure entry/exit delay zones & test siren
W16	**Troubleshooting Lab**	2 Hrs	1.0	- Bench loop diagnostics - Multimeter voltage tests - Fault repair verifications	- Installed trainer boards - Multimeter & probes - Fault diagnostic sheet	- Check Phase-Earth voltage using test lamp - Find short circuit fault inside junction box - Resolve induced IP clashes on camera net

3. Recommended Weekly Schedule Template

To maximize hands-on capability, each weekly topic is split across a 5-day progression. The daily 2-hour session is structured to minimize passive listening and maximize physical interaction with hardware.

Daily Session Time-Block Allocation

Within each 2-hour (120 minutes) daily session, manage time using the following split:

Phase	Duration	Focus Area	Key Activities
1. Concept & Demo	30 Minutes	Instructor-led briefing	Explaining core concepts, safety standards, and reviewing real-world wiring diagrams.
2. Active Lab Time	75 Minutes	Hands-on student execution	Wiring physical devices, configuring software settings, and running integration tests.
3. Review & Reset	15 Minutes	Verification & lab hygiene	Checking connections, backing up configurations, cleaning the workbench, and Q&A.

5-Day Weekly Rotation

Each weekly topic should follow this progression:

Day	Focus	Session breakdown	Expected Student Output
Day 1	**Concept & Wiring Topology**	â€¢ **30 mins:** Diagram explanation, spec sheets. â€¢ **75 mins:** Unboxing, identification, physical mounting on workbench. â€¢ **15 mins:** Component safety check.	Accurate block diagram drawn in lab book; components physically set up.
Day 2	**Configuration & Setup**	â€¢ **30 mins:** Software parameters & programming theory. â€¢ **75 mins:** Setting IPs, configuring software, testing initial controls. â€¢ **15 mins:** Configurations backed up.	Configured individual device accessible via control terminal.
Day 3	**Hardware Deployment & Cabling**	â€¢ **30 mins:** Termination standards (RJ45, RS485, Coaxial). â€¢ **75 mins:** Crimping, routing, powering through POE/SMPS. â€¢ **15 mins:** Multimeter voltage testing.	Fully wired and powered subsystem.
Day 4	**System Integration**	â€¢ **30 mins:** Relays, dry/wet contacts, trigger logic schemas. â€¢ **75 mins:** Interfacing subsystems (e.g. Video Door Phone triggering EM Lock). â€¢ **15 mins:** End-to-end flow validation.	Multi-device functional system executing logical commands.
Day 5	**Troubleshooting Lab & Assessment**	â€¢ **15 mins:** Fault-finding checklist methodology. â€¢ **90 mins:** Instructor-injected hardware/software fault resolution. â€¢ **15 mins:** Clean workbench reset.	Student successfully finds and repairs 2-3 faults.

4. Key Implementation Guidelines

IMPORTANT: **Theory to Practical Ratio (30:70)** Do not spend more than 35 minutes per day on slide presentations. Use physical devices on the workbench from Day 1.

TIP: **The Fault-Injection Method** The best way to build troubleshooting skills is for the instructor to secretly create problems (e.g., swapping Tx/Rx wires, setting static IP conflicts, cutting power supply channels) and having students resolve them using multimeters and software diagnostic tools.

> **NOTE:** **BOQ & Survey Portfolio** By the end of Module 4, every student should have built at least 3 custom proposals (Villa Smart Home, Office CCTV, Warehouse Security) to present as their portfolio.